

OUT LINES

- PHARYNGEAL APPARATUS

 - pharyngeal arch

 - pharyngeal pouch

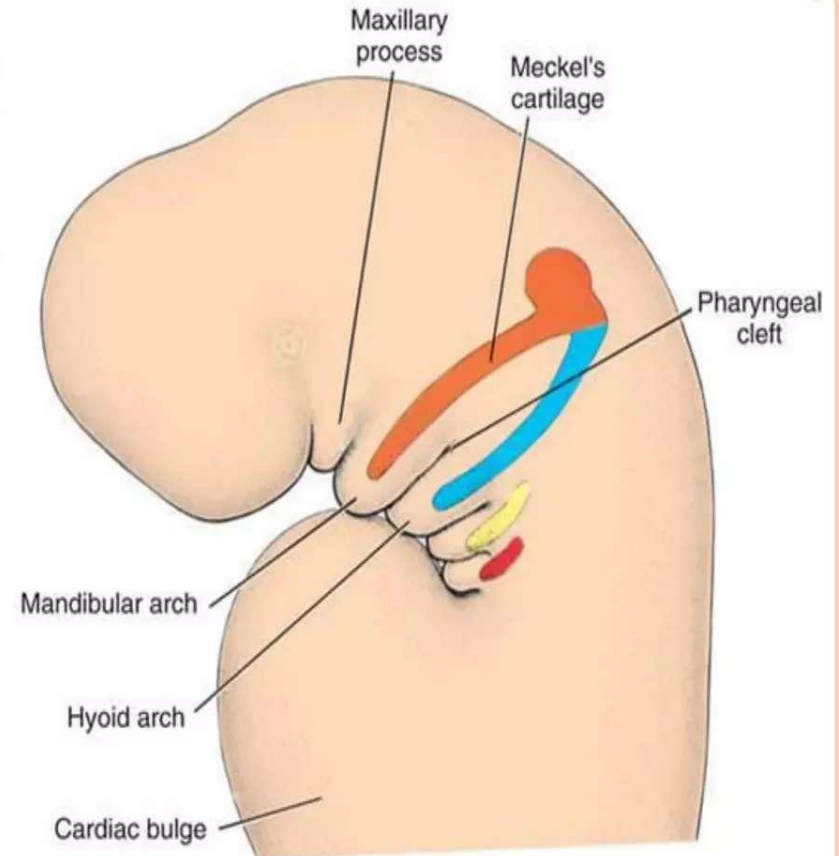
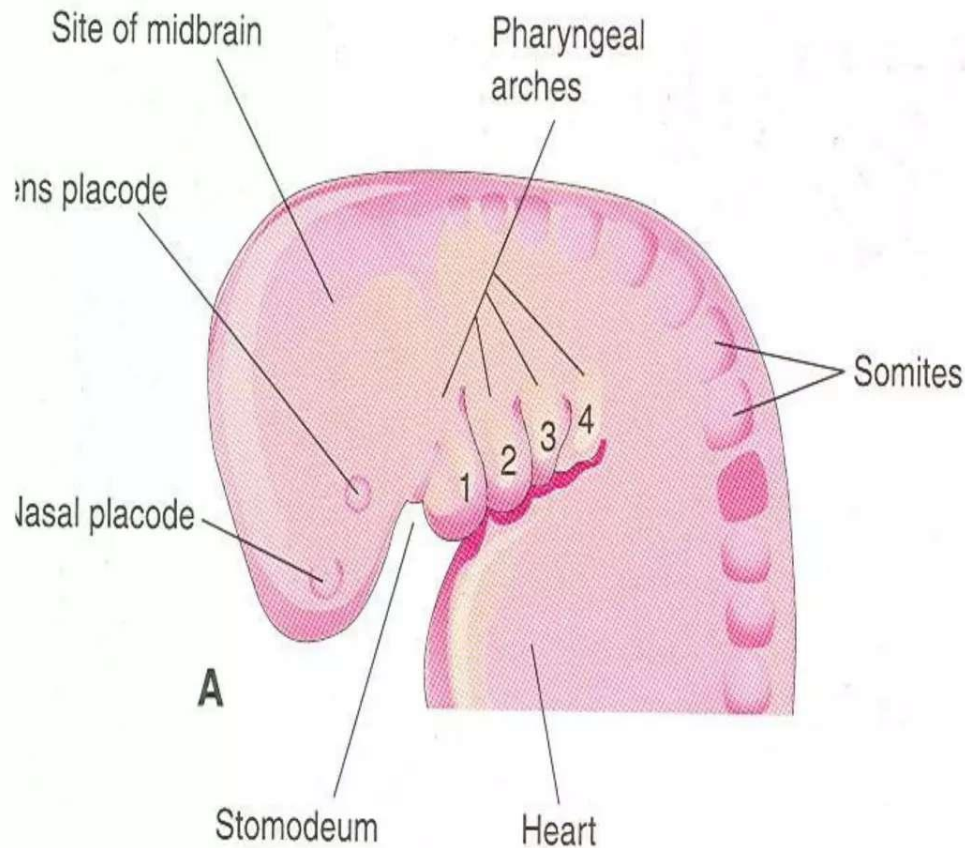
 - pharyngeal groove

 - pharyngeal membrane

- CLINICAL CORRELATIONS

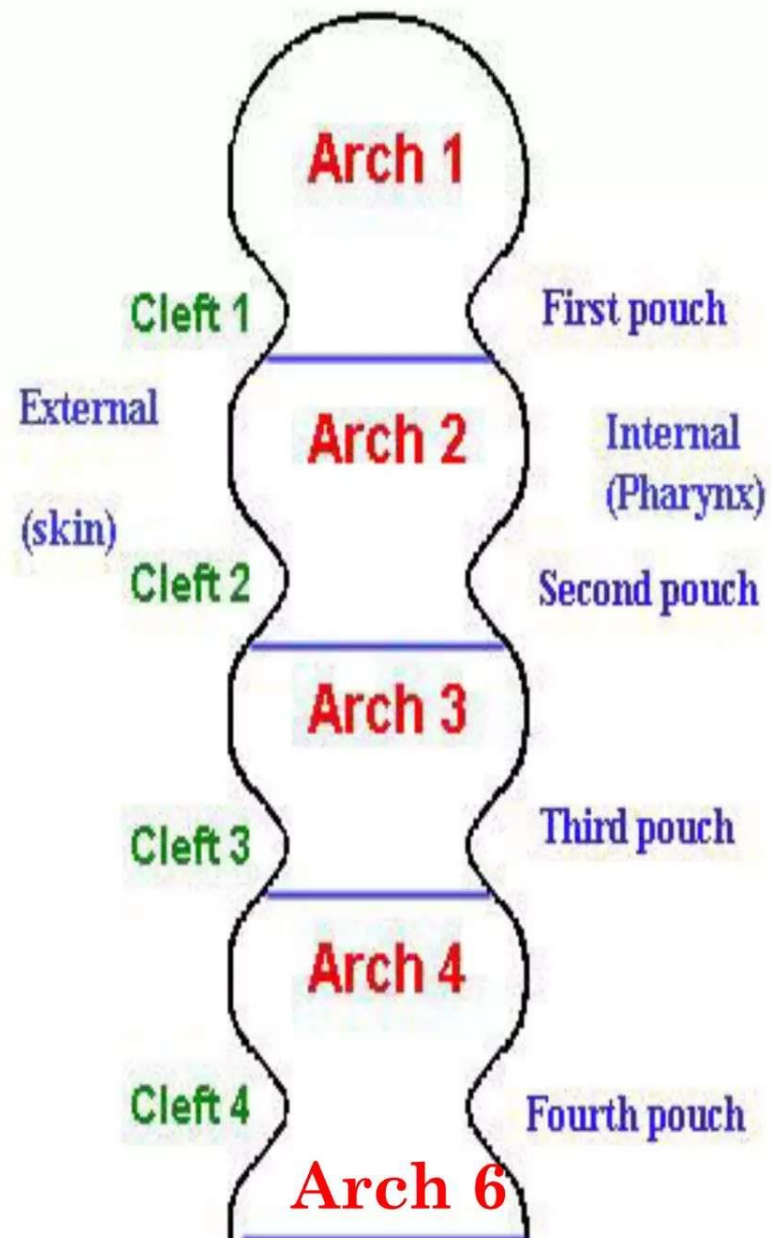
PHARYNGEAL / BRANCHIAL APPARATUS

4th week of intrauterine development



PHARYNGEAL APPARATUS

- The former uses of pharyngeal apparatus is branchial ,which means gills of fish
- A primordial pharyngeal (branchial) apparatus develops in human embryos . Consequently, the term pharyngeal arch is now used instead of branchial arch when describing the development of the head and neck regions of human embryos.



CONT..

- The first pair of pharyngeal arches, the primordial of the jaws, appears as surface elevations lateral to the developing pharynx.
- The pharyngeal arches are separated from each other by the pharyngeal grooves. Like the pharyngeal arches, the grooves are numbered in a craniocaudal sequence.

THE FIRST PHARYNGEAL ARCH

- Divide into:
 1. **Maxillary prominence** gives rise to
 - **The maxilla,**
 - **Zygomatic bone, and**
 - **Portion of the vomer.**
 2. **The mandibular prominence** forms the mandible.
 - The proximal mandibular prominence also forms **the squamous temporal bone**

CONT..

- The second pharyngeal arch (hyoid arch) contributes, along with parts of the third and fourth arches, to the formation of the hyoid bone.
- The pharyngeal arches support the lateral walls of the primordial pharynx, which is derived from the cranial part of the foregut.

A TYPICAL PHARYNGEAL ARCH CONTAINS

- A cartilaginous rod
- A muscular component
- A nerve
- A pharyngeal arch artery

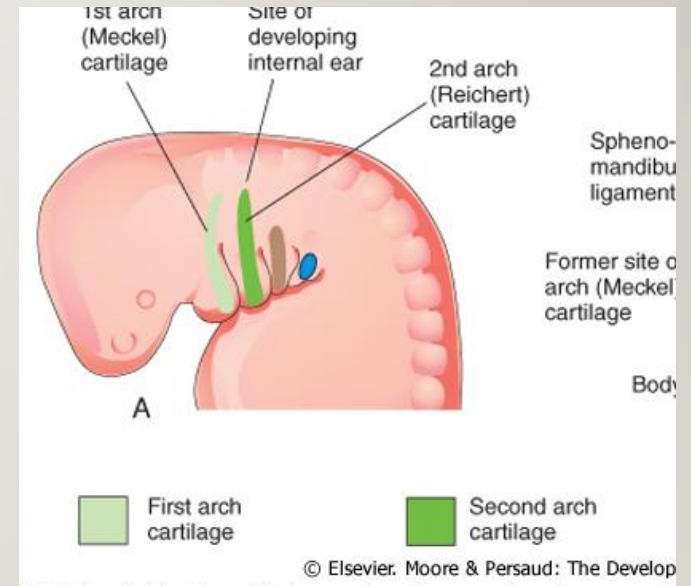
FATE OF THE PHARYNGEAL ARCHES

- The **pharyngeal arches** gives
the face,
nasal cavities,
mouth
larynx
pharynx, and neck .

DERIVATIVES OF THE PHARYNGEAL ARCH CARTILAGES

- The **first pharyngeal arch cartilage** (Meckel cartilage).

- I. Incus
- II. Malleus
- III. Anterior ligament of malleus
- IV. Sphenomandibular ligament
- V. Template for mandible



THE 2ND PHARYNGEAL ARCH CARTILAGES

- is called **Reichert cartilage**
- Give rise to
 - The *stapes of the middle ear*
 - the *styloid process of the temporal bone* .
 - The lesser cornu (Latin, horn) and
 - The superior part of the body of the hyoid bone

THE 3RD PHARYNGEAL CARTILAGES

- Give rise to:

1. Greater cornu

2. The inferior part of the body of the hyoid bone.

THE 4TH AND 6TH PHARYNGEAL ARCH CARTILAGES

- Gives rise to:
 - Laryngeal cartilages except epiglottis, which is derived from hypobronchial emeience.

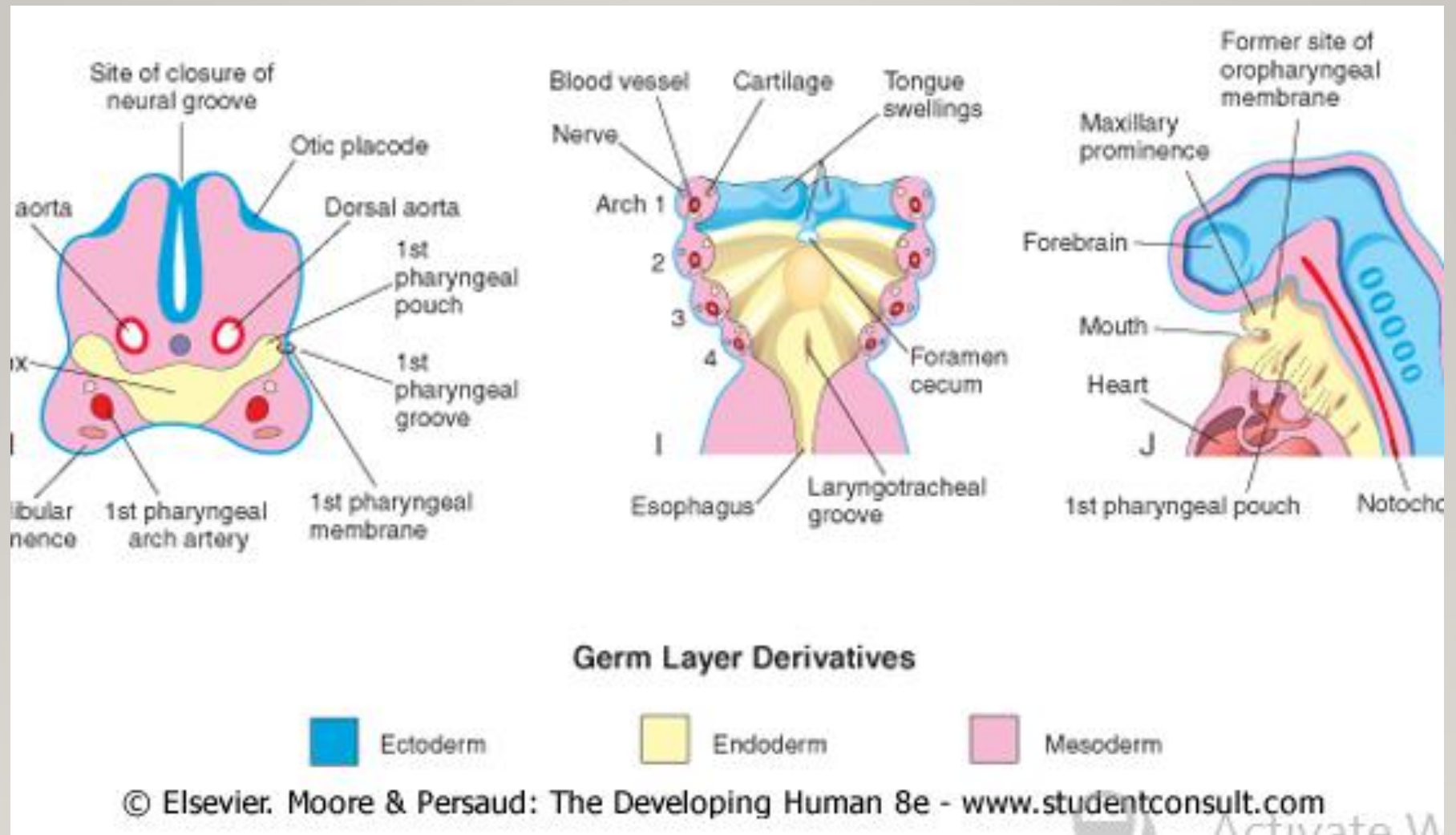


Fig 2 illustrating development of cartilages from pharyngeal arches

DERIVATIVES OF PHARYNGEAL ARCH MUSCLES

- The musculature of the *first pharyngeal arch* gives
 - i. Muscles of mastication
 - ii. Mylohyoid
 - iii. Anterior belly of digastrics
 - iv. Tensor tympani
 - v. Tensor veli palatini

CONT..

- The musculature of the *second pharyngeal arch* forms

I. Stapedius

II. Stylohyoid,

III. Posterior belly of digastric,

IV. Muscles of facial expression

The musculature of third pharyngeal arch give rise to:

Stylopharyngeus muscle.

CONT..

- The musculature of the *fourth pharyngeal arch* forms

I. The cricothyroid,

II. levator veli palatini, and

III. constrictors of the pharynx.

- The musculature of the *sixth pharyngeal arch* forms:

intrinsic muscles of the larynx

DERIVATIVES OF THE PHARYNGEAL ARCH NERVES

- The 1st pharyngeal arch gives:

Trigeminal nerve

- The 2nd pharyngeal arch

Facial nerve

- The 3rd pharyngeal arch

Glossopharyngeal nerve

CONT..

- The 4th and the 6th pharyngeal arch gives nerves of
 - I. superior branch of vagus nerve
 - II. Recurrent laryngeal nerve

DERIVATIVES PHARYNGEAL ARCH ARTERY

- The atreial component of 1st pharyngeal arch
 - Gives the terminal part of **maxillary artery**
- The 2nd pharyngeal arch disappear except embryological **hyoid and stapedia** arteries
- The atreal components of 3rd pharyngeal arch

Develops into: **.Common carotid artery and Internal carotid artery**

THE ARTERIAL COMPONENTS OF 4TH AND 6TH PHARYNGEAL ARCHES

- Those derived from the 4th aortic arch artery

1. Part of arch of aorta on the left side

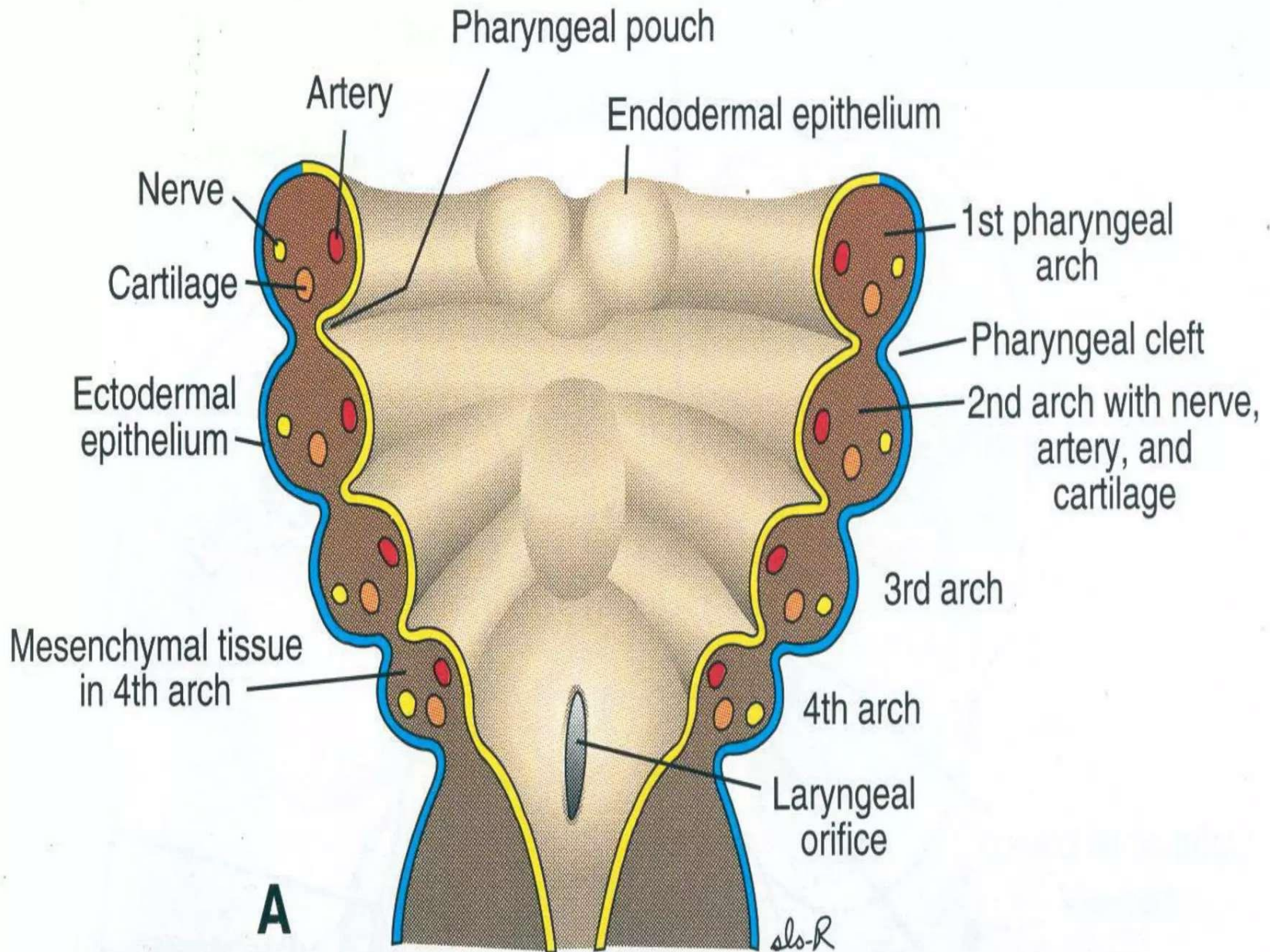
2. Proximal part of right subclavian artery on the right side

Those derived from 6th aortic arch artery

1. Right and left pulmonary artery

2. Ductus arteriosus from the left 6th aortic arch artery





THE FIRST PHARYNGEAL POUCH

The first pharyngeal pouch expands into an elongate **tubotympanic recess** with auditory vesicles it give rise to.

1. The tympanic membrane (eardrum).
2. The tympanic cavity and
3. The mastoid antrum.

THE PHARYNGEAL POUCH

- The second pharyngeal pouch remains as the **tonsillar sinus** or fossa .
- The endoderm of the second pouch proliferates and grows into the underlying mesenchyme.
- The central parts of these buds break down, forming **tonsillar crypts** (pitlike depressions).
- The pouch endoderm forms the surface epithelium and the lining of the tonsillar crypts.
- It differentiates into lymphoid tissue, which soon organizes into the **lymphatic nodules** of the palatine tonsil.

THE THIRD PHARYNGEAL POUCH

- The third pharyngeal pouch expands and develops a solid, dorsal bulbar part and a hollow, elongate ventral part .
- By the sixth week, the epithelium of each *dorsal bulbar part of the pouch* begins to differentiate into an inferior parathyroid gland.
- The epithelium of the elongate *ventral parts of the pouch* proliferates, obliterating their cavities. These come together in the median plane to form the **thymus**

THE FOURTH PHARYNGEAL POUCH

- The fourth pharyngeal pouch also expands into dorsal bulbar and elongate ventral parts .
- By the sixth week, each dorsal part develops into a **superior parathyroid gland**, which lies on the dorsal surface of the thyroid gland. Because the parathyroid glands derived from the third pouches accompany the thymus, they are in a more inferior position than the parathyroid glands derived from the fourth pouches .

Germ Layer Derivatives



Ectoderm



Mesoderm



Endoderm

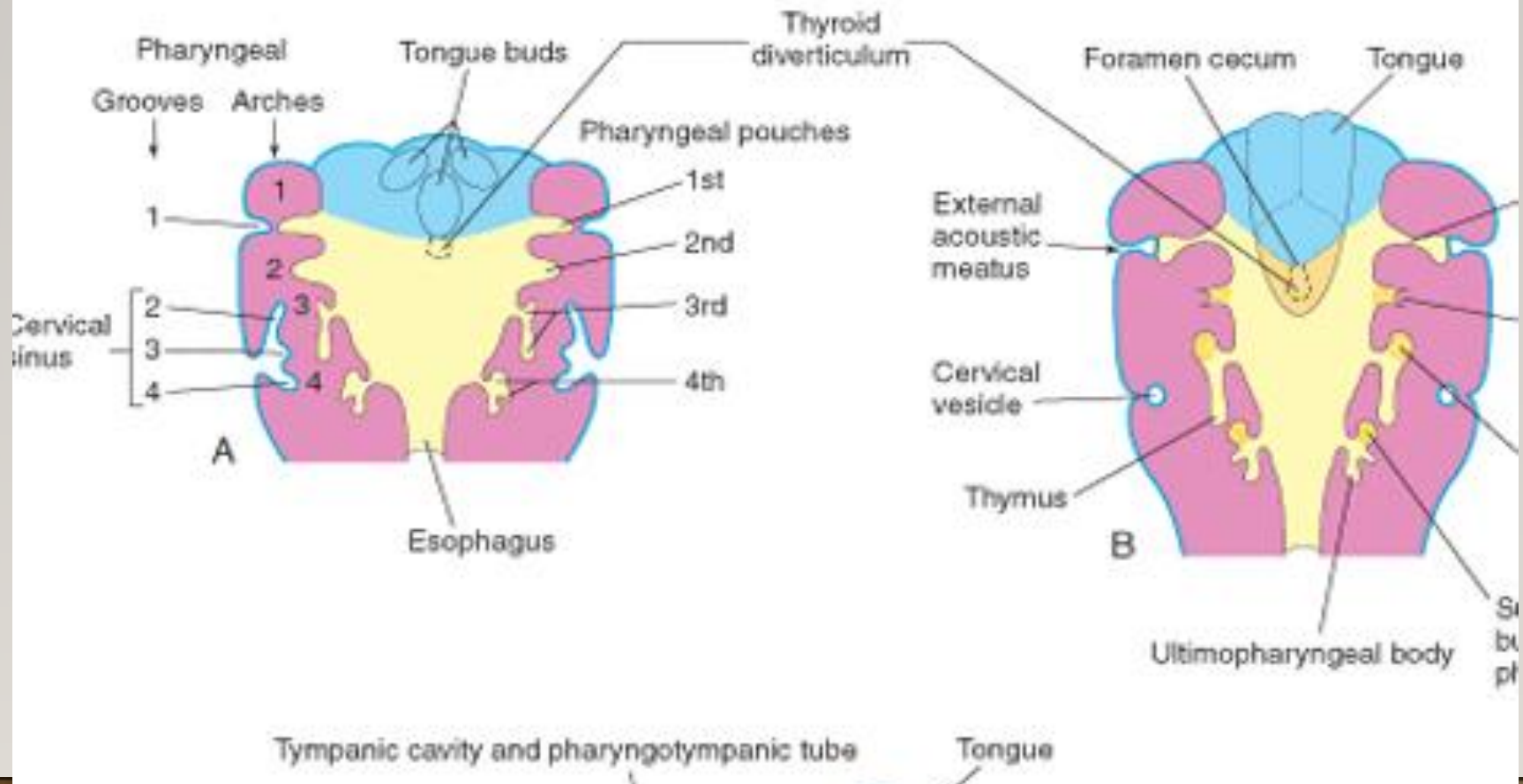


Fig 5 pharyngeal arches

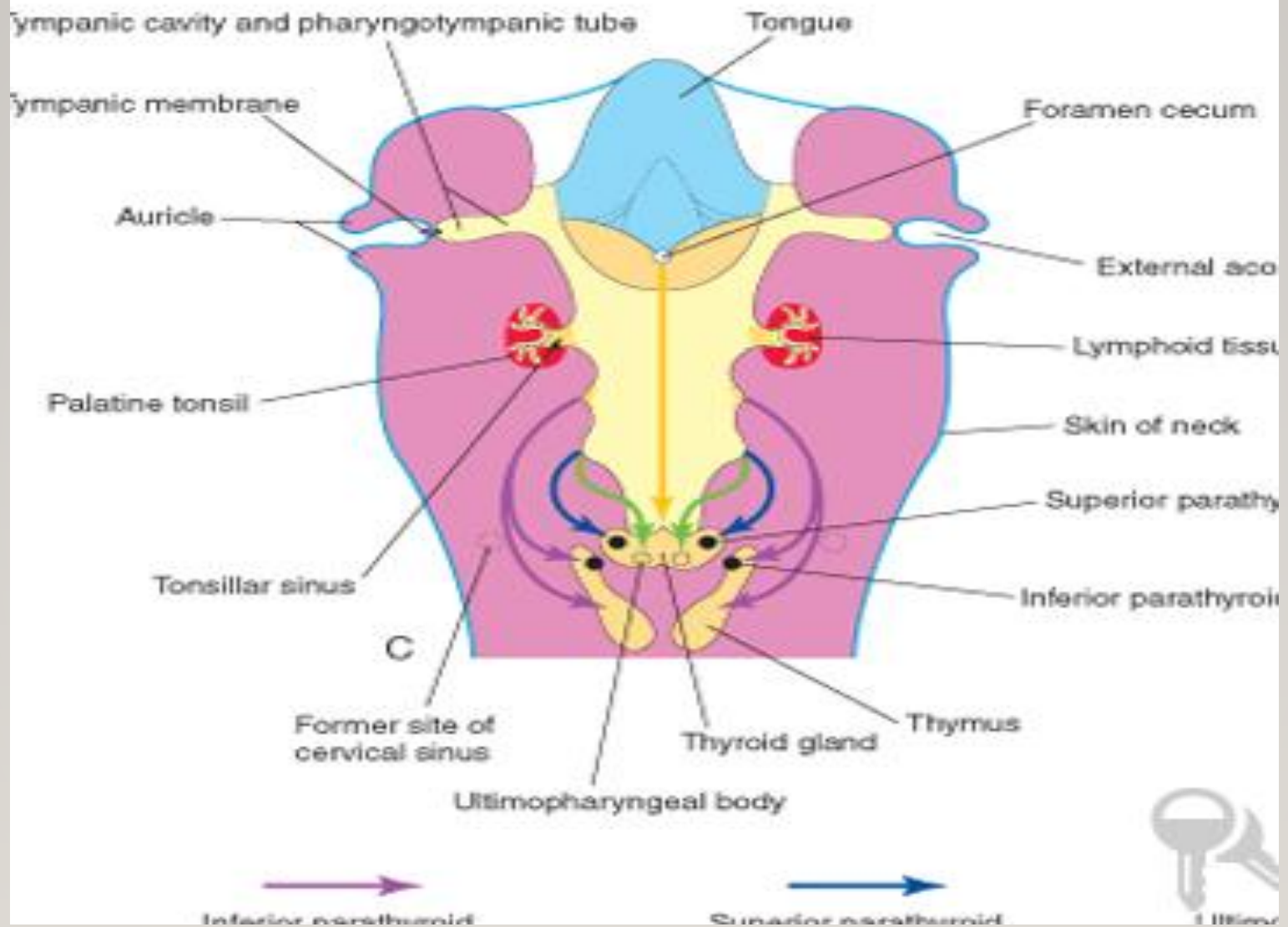
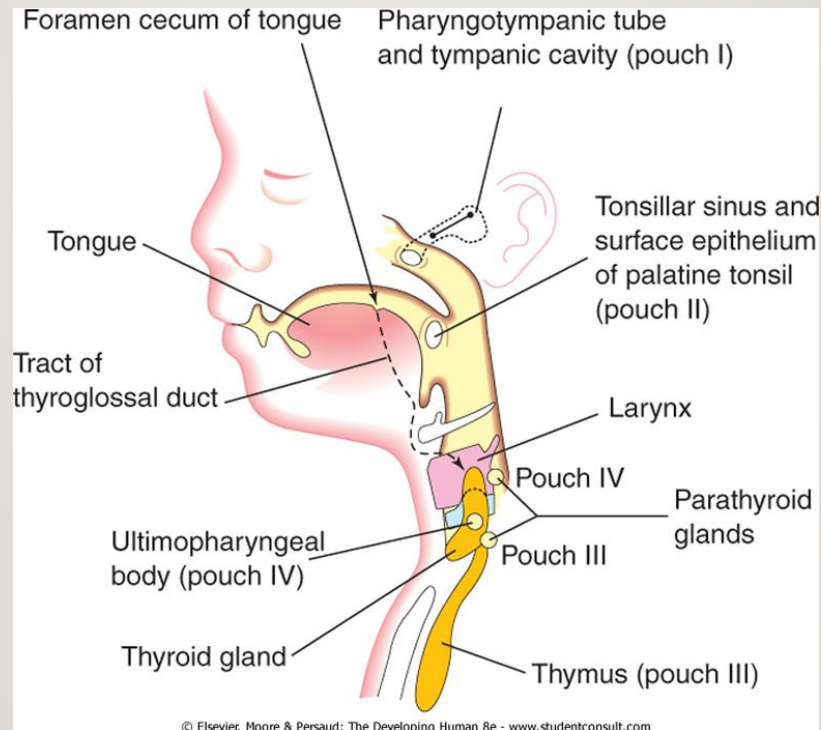


Fig 6 showing The Dev.t of thymus parathyroid and thyroid gland.



PHARYNGEAL GROOVES

- The head and neck regions of the human embryo exhibit four pharyngeal grooves (clefts) on each side during the fourth and fifth weeks
- The first pair persists as the **external acoustic meatus** or ear canals .
- The other grooves disappear.

PHARYNGEAL MEMBRANES

- The pharyngeal membranes appear in the floors of the pharyngeal grooves .
- Only one pair of membranes contributes to the formation of adult structures;
- The first **pharyngeal membrane**, gives rise to the **tympanic membrane** the rest disappear.

CLINICAL CORRELATION

- Atresia of Nasolacrimal Duct
- Bronchial Fistula an abnormal canal that opens internally into the tonsillar sinus and externally in the side of the neck is a branchial fistula. This canal results from persistence of parts of the second pharyngeal groove and second pharyngeal pouch.

CONT..

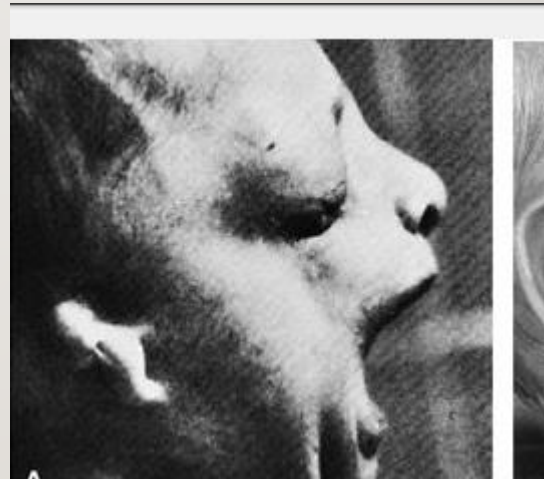
- **Bronchial Cysts**

Remnants of parts of the cervical sinus and/or the second pharyngeal groove may persist and form a spherical or elongate cyst



CONT..

- First Arch Syndrome
- **Treacher Collins syndrome** caused by an autosomal dominant gene due to under dev.t of zygomatic bone
- .



CONT..

- In **Pierre Robin syndrome** an autosomal recessive disorder, is associated with hypoplasia of the mandible, cleft palate, and defects of the eye and ear are present



CONT..

- **DiGeorge Syndrome** Infants with are born without a **thymus and parathyroid** glands and have defects in their cardiac outflow tracts.
- DiGeorge syndrome occurs because the third and fourth pharyngeal pouches fail to differentiate into the thymus and parathyroid glands



CONT..

- Ectopic Parathyroid Glands The location of the parathyroid glands is highly variable.
- They may be found anywhere near or within the thyroid gland or thymus.
- The superior glands are more constant in position than the inferior ones. Occasionally, an inferior parathyroid gland remains near the bifurcation of the common carotid artery

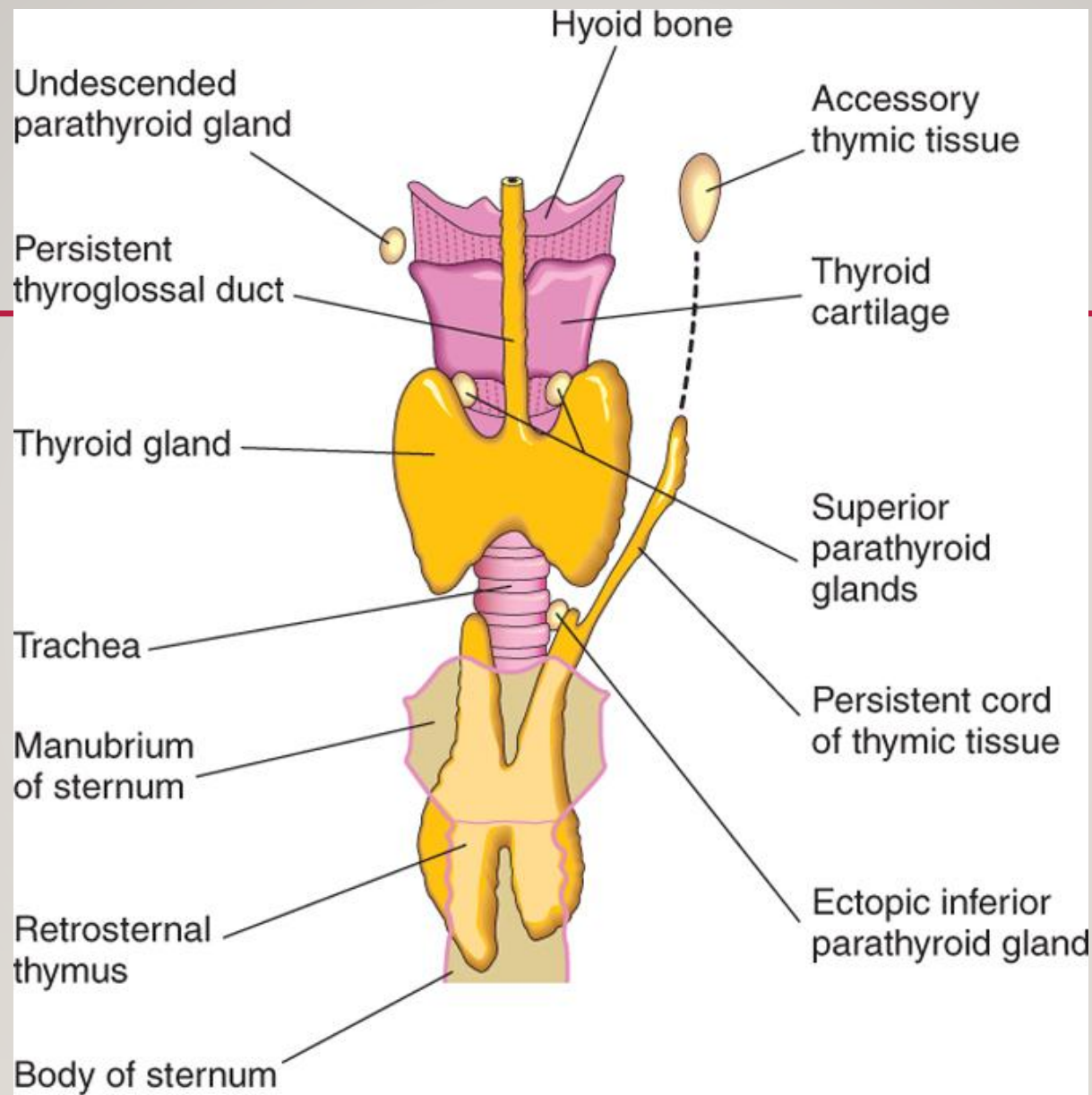


Fig 9 Congenital anomalies of thymus, parathyroid gland and thyroid gland

SUMMARY

- The primordial pharynx is bounded laterally by **pharyngeal arches**.
- Each arch consists of a core of mesenchyme covered externally by ectoderm and internally by endoderm. The original mesenchyme of each arch is derived from mesoderm; later, **neural crest cells** migrate into the arches and are the major source of their connective tissue components, including cartilage, bone, and ligaments in the oral and facial regions. Each pharyngeal arch contains an artery, a cartilage rod, a nerve, and a muscular component

CONT..

- Externally the pharyngeal arches are separated by pharyngeal grooves.
- Internally the arches are separated by evaginations of the pharynx-**pharyngeal pouches**.
- Where the ectoderm of a groove contacts the endoderm of a pouch, **pharyngeal membranes** are formed.

CONT..

- The **pharyngeal grooves** disappear except for the first pair, which persists as the external acoustic meatus. The pharyngeal membranes also disappear, except for the first pair, which becomes the tympanic membranes.
- The **first pharyngeal pouch** gives rise to the tympanic cavity, mastoid antrum, and pharyngotympanic tube.
- The **second pharyngeal pouch** is associated with the development of the palatine tonsil.

CONT..

- The **second pharyngeal pouch** is associated with the development of the palatine tonsil. The **thymus** is derived from the **third pair of pharyngeal pouches**, and the **parathyroid glands** are formed from the **third and fourth pairs of pharyngeal pouches**.

Quiz

- 1) Enumerate the derivatives of first and second PA
- 2) Explain Treacher Collins Syndrome